BUS314 Fall 2025
Midterm I Practice Questions and Answers

1) An S corporation earns \$9.10 per share before taxes. The corporate tax rate is 39%, the personal tax rate on dividends is 15%, and the personal tax rate on non-dividend income is 36%. What is the total amount of taxes paid if the company pays a \$5.00 dividend?
A) \$3.28
B) \$3.93
C) \$2.62
D) \$4.59
Answer: A
Explanation: \$9.10 × 36% = \$3.28

2) A C corporation earns \$8.30 per share before taxes. The corporate tax rate is 39%, the personal tax rate on dividends is 15%, and the personal tax rate on non-dividend income is 36%. What is the total amount of taxes paid if the company pays a \$6.00 dividend?
A) \$3.31
B) \$4.96
C) \$4.14
D) \$5.79
Answer: C
Explanation: Corporate tax = \$8.30 × 39% = \$3.24, Personal tax = \$6×15%=\$0.90.
Total tax = \$3.24 + \$0.90 = \$4.14

3) Which of the following people can not manage the operations of a firm in which they are part or full owners?
A) stockholders in S corporations
B) stockholders in C corporations
C) limited partners in a limited partnership
D) general partners in a limited partnership
Answer: C

4)

Luther Corporation
Consolidated Balance Sheet
December 31, 2006 and 2005 (in \$ millions)

Assets	**2006**	**2005**	**Liabilities and Stockholders' Equity**	**2006**	**2005**
Current Assets			*Current Liabilities*		
Cash	65.7	58.5	Accounts payable	87.7	73.5
Accounts receivable	54.4	39.6	Notes payable / short-term debt	9.6	9.6

Inventories	46.1	42.9	Current maturities of long-term debt	39.9	36.9
Other current assets	5.1	3.0	Other current liabilities	6.0	12.0
Total current assets	171.3	144.0	Total current liabilities	143.2	132.0
Long-Term Assets			*Long-Term Liabilities*		
Land	66.6	62.1	Long-term debt	237.7	168.9
Buildings	106.2	91.5	Capital lease obligations		
Equipment	119.3	99.6			
Less accumulated depreciation	(56.6)	(52.5)	Deferred taxes	22.8	22.2
Net property, plant, and equipment	235.5	200.7	Other long-term liabilities	---	---
Goodwill	60.0	--	Total long-term liabilities	260.5	191.1
Other long-term assets	63.0	42.0	Total liabilities	403.7	323.1
Total long-term assets	358.5	242.7	Stockholders' Equity	126.1	63.6
Total Assets	**529.8**	**386.7**	**Total liabilities and Stockholders' Equity**	**529.8**	**386.7**

Refer to the balance sheet above. Luther's quick ratio for 2006 is closest to ________.
A) 0.87
B) 1.75
C) 0.88
D) 1.31
Answer: A
Explanation: Quick ratio = (Current assets – Inventory) / Current liabilities
Quick ratio = ($171.3 – $46.1) / $143.2 = 0.87

5)

Luther Corporation
Consolidated Income Statement
Year ended December 31 (in $millions)

	2006	**2005**
Total sales	610.1	578.3
Cost of sales	-500.2	-481.9
Gross profit	109.9	96.4
Selling, general, and administrative expenses	-40.5	-39.0

Research and development	-24.6	-22.8
Depreciation and amortization	-3.6	-3.3
Operating income	41.2	31.3
Other income	--	--
Earnings before interest and taxes (EBIT)	41.2	31.3
Interest income (expense)	-25.1	-15.8
Pretax income	16.1	15.5
Taxes	-5.5	-5.3
Net income	10.6	10.2
Price per share	$16	$15
Sharing outstanding (millions)	10.0	8.1
Stock options outstanding (millions)	0.3	0.2
Stockholders' Equity	126.6	63.6
Total Liabilities and Stockholders' Equity	533.1	386.7

Refer to the income statement above. Assuming that Luther has no convertible bonds outstanding, then for the year ending December 31, 2006 Luther's diluted earnings per share are closest to ________.

A) $1.03

B) $0.51

C) $0.82

D) $1.23

Answer: A

Explanation: Diluted EPS = Net income / (Shares outstanding + Options contracts outstanding + Shares possible from convertible bonds outstanding)

= 10.6 / (10+0.3+0) =$1.03.

6) If the interest rate is 9%, the one-year discount factor is equal to ________.

A) 0.090

B) 1.090

C) 0.917

D) 0.981

Answer: C

Explanation: $1 / (1.09)^1 = 0.917$

7) Sara wants to have $600,000 in her savings account when she retires. How much must she put in the account now, if the account pays a fixed interest rate of 8%, to ensure that she has $600,000 in 20 years?

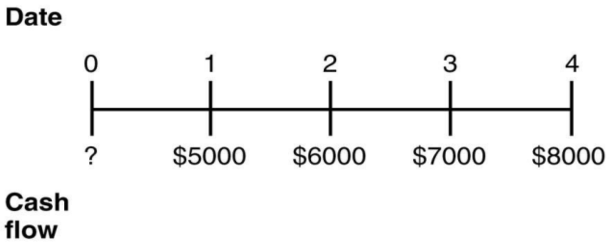
A) $128,729

B) $180,221

C) $231,712

D) $139,541
Answer: A
Explanation: N= 20，I/Y= 8, PMT =0, FV=600000, compute PV=-128,728.92

8) Consider the following timeline detailing a stream of cash flows:



If the current market rate of interest is 10%, then the present value (PV) of this stream of cash flows is closest to ________.
A) $10,114
B) $20,227
C) $24,272
D) $32,363
Answer: B
Explanation:
$PV = 5{,}000 / (1 + 0.1)^1 + 6{,}000 / (1 + 0.1)^2 + 7{,}000 / (1 + 0.1)^3 + 8{,}000 / (1 + 0.1)^4 = \$20{,}227.44$

9) A business promises to pay the investor of $6000 today for a payment of $1500 in one year's time, $3000 in two years' time, and $3000 in three years' time. What is the present value of this business opportunity if the interest rate is 6% per year?
A) $603.94
B) $301.97
C) $724.73
D) $966.30
Answer: A
Explanation: Calculate the NPV using CF keys: input $CF0 = -\$6000$, $CF1 = \$1500$, $CF2 = \$3000$, and $F2 = 2$ using $\text{interest} = 6\%$, which gives $NPV = \$603.9415$.

10) A perpetuity will pay $900 per year, starting five years after the perpetuity is purchased. What is the present value (PV) of this perpetuity on the date that it is purchased, given that the interest rate is 11%?
A) $2695
B) $4312
C) $5390
D) $3234
Answer: C
Explanation: The first step is to calculate the perpetuity price at t=4, which is C/r = 900/0.11=$8,181.82; the next step is to calculate its PV at t=0, which is 8181.82/(1+11%)^4= - 5389.6171.

11) Since your first birthday, your grandparents have been depositing $1200 into a savings account on every one of your birthdays. The account pays 6% interest annually. Immediately after your grandparents make the deposit on your 18th birthday, the amount of money in your savings account will be closest to ________.
A) $37,086.78
B) $22,252.07
C) $44,504.14
D) $51,921.49
Answer: A
Explanation: N = 18, I/Y = 6, PV = 0, PMT = -$1200, Compute FV = $37,086.78.

12) You are interested in purchasing a new automobile that costs $33,000. The dealership offers you a special financing rate of 9% APR (0.75% per month) for 60 months. Assuming that you do not make a down payment on the auto and you take the dealer's financing deal, then your monthly car payments would be closest to ________.
A) $548
B) $685
C) $959
D) $1096
Answer: B
Explanation: N = 60, I/Y = 0.75, PV = 33,000, FV = 0, Compute PMT = -$685.03.

13) Assume that you are 30 years old today, and that you are planning on retirement at age 65. You expect your salary to be $42,000 one year from now and you also expect your salary to increase at a rate of 5% per year as long as you work. To save for your retirement, you plan on making annual contributions to a retirement account. Your first contribution will be made on your 31st birthday and will be 8% of this year's salary. Likewise, you expect to deposit 8% of your salary each year until you reach age 65. Assume that the rate of interest is 9%.

The present value (PV) (at age 30) of your retirement savings is closest to ________.
A) $61,303
B) $30,652
C) $42,912
D) $67,433
Answer: A
Explanation: First deposit = 0.08 × $42,000 = $3360.00

$$\$3360.00 \times \frac{1}{0.09 - 0.05}\left(1 - \left(\frac{1 + 0.05}{(1 + 0.09)}\right)^{35}\right) = \$61,303$$

14) The effective annual rate (EAR) for a loan with a stated annual percentage rate

(APR) of 11% compounded quarterly is closest to ________.
A) 12.61%
B) 13.75%
C) 11.46%
D) 14.90%
Answer: C
Explanation: $EAR = (1 + APR / m)^m - 1 = (1 + 0.11/4)^4 - 1 = 0.1146$ or 11.46%

15) Consider the following investment alternatives:

Investment	APR	Compounding
A	6.9030%	Annual
B	6.6992%	Daily
C	6.7787%	Quarterly
D	6.7643%	Monthly

Which alternative offers you the lowest effective rate of return?
A) Investment A
B) Investment B
C) Investment C
D) Investment D
Answer: A
Explanation: $EAR\ (A) = (1 + APR / m)^m - 1 = (1 + 0.069030/1)^1 - 1 = 6.9030\%$
$EAR\ (B) = (1 + APR / m)^m - 1 = (1 + 0.066992 /365)^{365} - 1 = 6.9280\%$
$EAR\ (C) = (1 + APR / m)^m - 1 = (1 + 0.067787/4)^4 - 1 = 6.9530\%$
$EAR\ (D) = (1 + APR / m)^m - 1 = (1 + 0.067643 /12)^{12} - 1 = 6.9780\%$

16) You are considering purchasing a new home. You will need to borrow $290,000 to purchase the home. A mortgage company offers you a 20-year fixed rate mortgage (240 months) at 12% APR. If you borrow the money from this mortgage company, your monthly mortgage payment will be closest to ________.
A) $2554
B) $4470
C) $3193
D) $5109
Answer: C
Explanation: N = 240, I/Y = 1, PV = 290,000, FV = 0, Compute PMT = -$3193.15.

17) Salvatore has the opportunity to invest in a scheme which will pay $5000 at the end of each of the next 5 years. He must invest $10,000 at the start of the first year and an additional $10,000 at the end of the first year. What is the net value of this investment if the interest rate is 3%?
A) -$3189.80
B) -$5907.57

C) 5907.57
D) $3189.80
Answer: D
Explanation: The first step is to calculate the investment in PV terms:
PV of investment = 10,000 + 10,000 /(1+3%) = 19,708.7379.
The second step is to use CF function to compute NPV:
CF0= -19,708.74 $CF1 = \$5000,\ \ F1 = 5;$ using interest $rate = 3\%,\ \ NPV = \$3189.80.$

18) A truck costing $111,000 is paid off in monthly installments (to be paid at the end of each month) over four years with 8.10% APR. After three years the owner wishes to sell the truck. What is the closest amount from the following list that he needs to pay on his loan before he can sell the truck?
A) $24,956
B) $37,434
C) $31,195
D) $43,673
Answer: C
Explanation: The first step is to calculate the monthly payment
N= 48, I/Y= 8.1/12, PV=111,000, FV=0, compute PMT = -2,715.0478.
The second step is to use that monthly payment to calculate the present value (PV) of 12 months remaining payment keeping the interest rate unchanged.
N = 12, I/Y = 0.675, PMT = - 2,715.0478, FV =0, compute PV = 31,195

19) In 2007, interest rates were about 4.5% and inflation was about 2.8%. What was the real interest rate in 2007?
A) 1.58%
B) 1.61%
C) 1.62%
D) 1.65%
Answer: D
Explanation: real rate = 1.045 / 1.028 -1 = 1.0165 – 1 = 1.65%

20) Elinore is asked to invest $5100 in a friend's business with the promise that the friend will repay $5610 in one year. Elinore finds her best alternative to this investment, with similar risk, is one that will pay her $5508 in one year. What is the opportunity cost of capital in this case?
A) 7%
B) 8%
C) 9%
D) 10%
Answer: B
Explanation: $5508 - $5100 = $408; $408 / $5100 = 0.08 or 8%